MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL 2026

ASSIGNMENT 2

Due date: September 28 (Monday), 22:00 pm.

(1) Let (a_n) be a sequence of positive integers. Define

$$x_1 = a_1, \quad x_2 = a_1 + \frac{1}{a_2}, \quad x_3 = a_1 + \frac{1}{a_2 + \frac{1}{a_3}}, \quad x_4 = a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}},$$

and so on.

(a) Show that the subsequences (x_{2n}) and (x_{2n-1}) converge.

(b) Suppose that $a_n = 1$ for all n. Determine whether (x_n) converges. Find $\lim_{n\to\infty} x_n$, if it exists. (Hint: You may use [BS, Section 3.4, Exercise 6], or see the discussion from the lecture on September 17.)

(2) Let $A \subset \mathbb{R}$. An element $x \in \mathbb{R}$ is said to be a cluster point of A if there exists a sequence (x_n) in A such that (x_n) converges to x. Note that x is not necessarily an element of A.

(a) Show that an element $x \in \mathbb{R}$ is a cluster point of A if and only if there exists a sequence (x_n) in A such that

$$|x_n - x| < \frac{1}{n}$$

for all n.

(b) Let $A \subset \mathbb{R}$ be nonempty and bounded. Let $\mathcal{M}$ be the set of all cluster points of A. Show that the maximum of $\mathcal{M}$ exists.

Reference Book: [BS] Robert G. Bartle and Donald R. Sherbert, *Introduction to Real Analysis*, Wiley, 2011, Fourth Edition.